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A Synthesis of Factors Related to Trends in Abundance and Demography of Alaska Chinook Salmon (*Oncorhynchus tshawytscha*, Salmonidae): Implications for Research, Management, and Policy

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ABSTRACT

Across Alaska, there have been synchronous declines in the abundance, mean age, and size of Chinook salmon (*Oncorhynchus tshawytscha*, Salmonidae), a species with immense social and ecological importance. The decline of Chinook salmon, and regulations addressing declines, have drastically impacted people who rely on Alaska's fisheries. Despite their importance, the drivers causing Chinook salmon declines are still not well understood. Therefore, to improve our understanding of the likely drivers of these large-scale synchronous declines, we synthesised the peer-reviewed Chinook salmon literature published between 2010 and 2024 using a weight-of-evidence approach. We identified 94 variables used to explain the trends in abundance and demographics of Chinook salmon. Each variable's association with trends in Chinook salmon was individually assessed and integrated using a standardised method. Results of this synthesis suggest the drivers for large-scale synchronous declines have occurred in the North Pacific Ocean. Specifically, we found marine predation, size selective harvest, and pink salmon (*Oncorhynchus gorbuscha*) abundance are the factors most strongly associated with reduced size and age. Results also suggest ocean productivity—specifically winter sea surface temperature and the North Pacific Gyre Oscillation—are the most frequently identified predictors of regional Chinook salmon abundance. In addition to addressing freshwater factors like adult heat stress and juvenile predation, we suggest research, policy, and management focus on how to better understand and address the interactions between ocean productivity, marine predation, size selective harvest, and pink salmon abundance to improve the condition of Chinook salmon, and the people and ecosystems benefiting from them.

1 | Introduction

Rigorous, systematic, and comprehensive reviews of primary research are essential for distilling large bodies of literature into an objective and shared understanding (Haddaway et al. 2015). These reviews are especially imperative for complex social–ecological systems that are studied through diverse disciplines and methods (McGinnis and Ostrom 2014; Pullin and Stewart 2006). Here, we present a weight-of-evidence synthesis of data, research, and knowledge to help create a shared understanding of the factors associated with the widespread declines of Alaska Chinook Salmon (*Oncorhynchus tshawytscha*, Salmonidae)—a vitally important species with immense social and ecological value.

The need to understand what is driving Chinook salmon dynamics is especially urgent, as many populations have experienced an abrupt shift to chronically low productivity and abundance (Atlas et al. 2023; ADFG – DCF 2022; Joint Technical Committee of the Yukon River U.S./Canada Panel 2021). However, it is not just the declining abundance of Chinook salmon causing concern. Alaska Chinook salmon are declining in size-at-age and age at maturity, resulting in an overall decline in mean body size. These trends are being exhibited synchronously at a large geographic scale despite differences in habitat, harvest, and life history (Oke et al. 2020; Ohlberger et al. 2016, 2018; Lewis et al. 2015).

Declines in abundance and mean body size have a range of social and ecological implications, like decreased fecundity and a weakened portfolio effect (Staton et al. 2021; Ohlberger et al. 2020; Dorner et al. 2018). A reduced portfolio effect increases a species' risk to adverse events across space and time (Carvalho et al. 2023). In addition, body size declines have also reduced nutrient transport (−28%), fisheries value (−21%), and meals for subsistence (−26%) (Oke et al. 2020). Declines also increase regulations, limit access, and increase conflict (Lamborn et al. 2023; Loring 2016; Harrison and Loring 2014). Furthermore, restricting access to fisheries in Alaska threatens cultures, livelihoods, well-being, and the sovereignty of rural and Indigenous communities (Connoy et al. 2024; Carothers et al. 2021). Recently, petitions have been made for listing some Chinook salmon populations in Alaska under the U.S. Endangered Species Act of 1973, which has many complex and potentially unforeseen implications.

Although the impacts from these declines are obvious and acute, the processes driving the declines of Chinook salmon remain poorly understood. Using Ostrom's (2009) social–ecological framework, Lamborn et al. (2023) identified that it is the nature of salmon (migratory) and their habitat (large and unpredictable) that leads to uncertainty about effective management strategies. This uncertainty, especially when there is conflict among large and diverse sets of resource users, leads to many possible, and often conflicting, management interventions. Uncertainty and conflict ultimately hinder the governance system in [making decisions](#) toward sustainability. Resource users, broader invested parties, and the governance system “must have sufficient information available to them to allow for the creation of effective management” (Singleton 2000, 4).

The purpose of this synthesis is to provide clarity and facilitate the development of a shared understanding of the complex processes affecting the culturally and economically vital Chinook salmon. We intend the review to increase the likelihood that resource users and the governance system can move collectively toward a course of action aimed at social and ecological sustainability (Ostrom 2009; Pullin and Stewart 2006). Having a clear understanding of what is related to, and thus may be causing, undesirable conditions (i.e., moving a system away from sustainability) also reduces the likelihood of research, policy, and management focusing on insignificant or marginally significant components of the fishery, rather than the most probable root cause(s) (Welch et al. 2021). Misguided efforts can unnecessarily stifle economic activity, reduce wellbeing, increase conflict, and produce other unintended and undesirable outcomes, all the while delaying the implementation of effective management. Through this synthesis, our goal is to help improve our shared understanding of Alaska Chinook salmon to benefit research, management, and policy moving forward.

2 | Methods

2.1 | Overview

Our objective was to compile, review, and synthesise the literature relevant to Chinook salmon populations along the Alaskan coast of North America. Specifically, we were looking for literature focused on: (1) trends occurring among Chinook Salmon populations; and (2) information used to explain those trends.

For this systematic and integrative review, we took a hybrid approach following the processes developed by Pullin and Stewart (2006) and Torracco (2005). In adherence to these authors' recommendations, we have structured our methods into the following sections: (1) Literature search; (2) Literature selection; (3) Literature review; and (4) Literature analysis.

To present the findings of this synthesis, we plotted each independent variable used to explain the trends exhibited by Chinook salmon by the number of times each variable was studied and the strength of evidence supporting that variable's ability to explain trends in Chinook salmon abundance and size. Two matrices were created: *freshwater* (including both juvenile and adult life stages) and *ocean*. These matrices display the body of Alaskan Chinook salmon literature, making it easy to identify what variables have been studied, how many times they have been studied, and the average weight of evidence of each variable.

2.2 | Literature Search

As discussed by Pullin and Stewart (2006), we took a high-sensitivity and low-specificity approach to our literature search to reduce bias and increase repeatability. This approach is necessary for a species like Chinook salmon because the research spans a diverse range of topics, geography, life history, and methodological approaches. One drawback of this approach is that it returns a high number of articles that need to be manually filtered for relevance.

To cast this large literature-catching net, we used Scopus to conduct a title, abstract, and keyword search using the following query: ((Chinook OR tshawytscha) AND (Alaska OR {North Pacific} OR Bering OR AYK OR Yukon OR {Bristol Bay} OR Kenai OR {Cook Inlet})) AND PUBYEAR > 2009 AND PUBYEAR < 2025. The review was restricted to research conducted during or after 2010 because the primary trends we were interested in investigating have become acutely present within the last decade, and most studies using Chinook trend data date back decades.

2.3 | Literature Selection

Our Scopus search returned 240 documents published between 2010 and 2024. The titles and abstracts of these were manually screened by two of this paper's authors. All relevant publications were entered into a spreadsheet for a full text review (the authors, date of publication, title, and link were entered) (Data S1). Of these 240 articles, 104 were saved. During the full-text reading, articles were divided into two categories: (1) articles solely identifying trends; and (2) articles attempting to explain observed trends. We identified 29 “trend” articles and 51 “explanation” articles, and 25 articles were excluded from analysis. Papers were selected based on the following criteria:

- Publications must have been published in English between 2010 and the time of our review.
- The geographic range of the study must include, in part, the marine environment around Alaska or freshwater salmon habitat within Alaska.
- Publications must have some relevance or discussion of the 10 metrics of interest (i.e., dependent variables): abundance, age structure, age-at-maturity, fecundity, genetic diversity, size-at-age, sex ratio, survival, productivity, and run size (Table 1).
- *Explanation* articles must have tested at least one independent variable as a potential driver of change in a dependent variable related to Chinook salmon life-history, population stability, or fish health.

We excluded publications focused primarily on the behaviour or ecological niche occupied by Chinook salmon lacking commentary on broader population-scale impacts or effects. We also excluded research on Chinook salmon originating in Russian or Japanese freshwater environments with marine ranges extending into the Bering Sea, as these publications rarely included data discernibly relevant to Alaskan Chinook salmon populations. Lastly, papers primarily focused on the social impacts of Chinook salmon decline, while significant and important, were not included in this synthesis.

2.4 | Literature Review

To assess trends in Chinook salmon populations, we reviewed 28 studies that documented observed trends without analysing causal factors and 51 studies that explicitly investigated the drivers of these trends. For the first group, we recorded the

TABLE 1 | Definitions for metrics used to assess change in Chinook populations.

Metrics	Definitions
Abundance	Number of individuals at a given life stage and/or geography
Age structure	Average age or age composition of a population, cohort, return, spawners, etc.
Age-at-maturity	Age at which individuals become sexually mature
Fecundity	Number of eggs produced by female salmon
Genetic diversity	Prevalence of specific genetic material possibly impacted by population bottlenecks, habitat availability, or hatchery supplementation
Size-at-age	Average size of individuals in a population at a defined age
Sex ratio	Ratio of females to males in a population
Survival	The number of individuals surviving over a defined timeframe
Productivity	Recruits-per-spawner, or the number of offspring produced by a cohort
Run size	Number of individuals returning to a river to spawn in a given calendar year

study location, the time span of the data, and the reported biological changes, including shifts in size-at-age, age structure, fecundity, sex ratio, survival, and overall run size. For the second group, we systematically extracted key methodological components, including the study location, the life stage of the salmon examined, and the dependent and independent variables assessed.

Following best practices in systematic literature reviews, we adopted a structured weight-of-evidence approach to evaluate and integrate findings from this diverse body of research. The weight-of-evidence approach is widely applied in environmental and ecological risk assessment as a transparent and repeatable method for synthesising multiple lines of evidence (Linkov et al. 2009). Weight-of-evidence approaches can range from qualitative assessments that rely on expert judgement to quantitative approaches that integrate statistical measures and ranking systems. In this study, we employed a quantitative weight-of-evidence approach based on a structured scoring system that incorporated statistical significance, effect size, and geographic consistency as key evaluation criteria.

Each independent variable identified in the 51 studies investigating the drivers of Chinook salmon trends literature was systematically ranked according to its relationship with the dependent variable. Variables demonstrating strong statistical significance, substantial effect sizes, and consistency across multiple

populations and geographic regions were assigned the highest rankings, while those with weaker statistical support or limited generalisability were ranked accordingly. In this assessment, each independent variable was ranked by its association with the dependent variable using a scale that ranged from 2 = strong positive evidence; 1 = moderate positive evidence; 0 = no evidence; -1 = moderate negative evidence; and -2 = strong negative evidence (the structured protocol is detailed in the Data S2). To ensure consistency and reproducibility, two independent reviewers assigned scores, and any discrepancies were resolved through consensus discussions.

Given the diversity of research designs across the studies reviewed, a direct meta-analysis was not feasible. The methodologies ranged from hierarchical Bayesian models to observational field studies and controlled laboratory experiments, making direct statistical aggregation inappropriate. Instead, we applied a structured weight-of-evidence synthesis that allowed for the integration of different data types and methodological approaches while maintaining a quantitative assessment method. This approach ensured that findings were evaluated systematically and transparently, reducing the potential for subjective interpretation.

To illustrate how this weight-of-evidence approach was applied, we provide an example from the literature on adult migration heat stress, a factor frequently cited as influencing Chinook salmon productivity. In one study, Murdoch et al. (2024) used a hierarchical Bayesian stock-recruitment model to assess the effects of multiple independent variables on recruits-per-spawner in Yukon River Chinook salmon. Their analysis found a significant negative relationship between adult heat stress and productivity, with a posterior probability distribution indicating a strong statistical effect. Given the robustness of the modelling approach, the significance of the results, and the consistency of findings across multiple populations, adult heat stress in this study was assigned the highest possible ranking, indicating strong negative evidence (-2; high water temperature = reduced productivity).

In contrast, von Biela et al. (2022) compiled field observations of premature mortality in Pacific salmon during the 2019 Alaska drought. Their study identified river discharge and water temperature as key variables influencing survival. Reduced river discharge appeared to have a moderate negative effect on survival, and elevated water temperature was associated with increased mortality, though the effect on Chinook salmon varied across geographic regions. Due to the observational nature of the study and the variability in its findings across Alaska, river discharge was assigned a moderate positive ranking (1; higher discharge = improved survival), while water temperature was ranked as moderate negative evidence (-1; high water temperature = reduced survival).

A third study, Bowen et al. (2020), conducted a controlled experiment examining the physiological response of Yukon River Chinook salmon to varying water temperatures. Their results indicated that while no significant physiological stress was observed at 14°C or 18°C, exposure to 21°C resulted in high

physiological stress and mortality. Since the Yukon River only occasionally reaches or exceeds this temperature threshold, the findings were considered relevant but not broadly generalisable. As a result, the evidence from this study was assigned a moderate negative ranking (-1), reflecting its importance in specific contexts but its limited applicability to broader population trends.

Many papers examined multiple independent variables across multiple Chinook salmon populations or regions. If an analysis used 12 independent variables across two populations/regions independently, we assessed the evidence for all 12 variables independently. However, if the analysis of the populations/regions was not independent (e.g., Ohlberger et al. 2016), then we accounted for that in our assessment (e.g., downgrading significance from 2 to 1 because of inconsistent results across regions). In addition to ranking each independent variable, we also wrote a brief description for variables providing moderate to strong evidence (e.g., 2 (Survival increased with higher winter sea surface temperature in the Bering Sea)). This increased the resolution and more accurately reflects the specificity of results from the literature. Therefore, each analysis of a population/region/life stage was recorded in a separate row of the spreadsheet, including the location, Chinook salmon life stage, and dependent and independent variables. This spreadsheet is available in the Data S1.

By applying this structured weight-of-evidence approach, we ensured that our synthesis of the literature was both systematic and transparent. The ranking system provided a quantitative basis for evaluating the strength of evidence across studies utilising diverse methodological approaches, aligning with best practices in environmental assessment. In doing so, we followed the recommendations of Linkov et al. (2009), who emphasise the need for structured data integration techniques to improve the transparency and repeatability of ecological risk assessments. Our approach allowed for the meaningful comparison of findings from different research methodologies while maintaining the scientific rigour necessary for robust conclusions.

2.5 | Literature Analysis

After completing the spreadsheet, we calculated the mean significance for each independent variable, tallied the number of times it was studied, and plotted the independent variables in one of two matrices: one for the freshwater environment (for both juvenile and adult life stages) and one for the marine environment. The number of times independent variables were studied was plotted on the x-axis, and the average weight of evidence was plotted on the y-axis. We also differentiated which independent variables were used to explore trends associated with Chinook salmon abundance (i.e., numerical abundance, population productivity, recruits-per-spawner, or survival) and size (i.e., age structure, age-at-maturity, average age, average size, body size, fecundity, growth, maturation, and size-at-age). Independent variables with an average evidence strength of "0" were excluded from the matrices but are included in the Data S3 and S4.

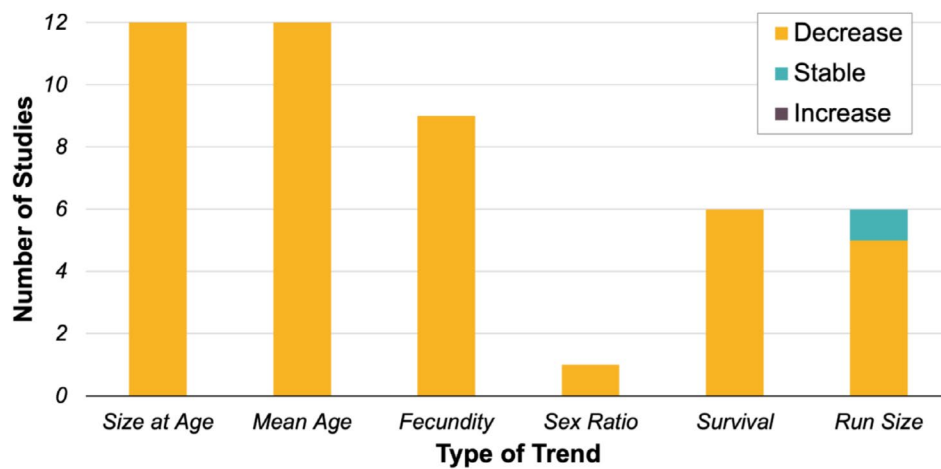


FIGURE 1 | Trends identified in the literature.

3 | Results

3.1 | Literature and Data

We identified 80 separate peer-reviewed studies focused on the trends of Chinook salmon in Alaska published between 2010 and 2024. Twenty-nine of these studies only identified trends occurring within Chinook salmon populations, but did not use any form of analysis to identify variables potentially associated with those trends. These ‘trend’ studies focused on the types of changes occurring, and geographic scope and synchrony of those changes. The other 51 studies (‘explanation’ studies) used some form of analysis to identify associations between observed trends and social (e.g., harvest) and/or environmental (e.g., temperature) variables.

The number of studies examining Chinook salmon trends increased in 2015, indicating the topic’s growing importance. In these studies, the earliest Chinook salmon population data dates to 1925, and the most recent dataset used ended in 2021. The mean start year of Chinook salmon datasets used in these studies is 1973, and the mean end year is 2014 (many studies used brood year).

3.2 | Chinook Salmon Trends in Alaska

The two most common trends identified in the literature were declining size-at-age and mean age (Figure 1), and both trends have been observed with a high degree of synchrony across a wide geography (Oke et al. 2020; Ohlberger et al. 2018; Lewis et al. 2015). The declines in size-at-age and mean age are also associated with reduced fecundity (i.e., smaller females have fewer eggs and less egg mass) (Ohlberger et al. 2020; Oke et al. 2020; Dorner et al. 2018). One study identified a negative trend in sex ratio, meaning returning Chinook salmon cohorts are composed of a higher proportion of males to females (Staton et al. 2021). Six studies also identified decreased survival (Brown et al. 2020; Welch et al. 2021). Lastly, five studies identified a decline in run size; however, many studies state run size is highly variable and there have only been slight declines when compared to the long-term average (Brown et al. 2020; Cunningham et al. 2018; Ohlberger et al. 2016). Taken together, the weight of evidence

strongly suggests that across Alaska we are in a period where survival rates are declining, and there are fewer, smaller, and younger Chinook salmon returning, which reduces the future reproductive potential of populations (e.g., fecundity).

3.3 | Variables Used to Explain Trends in Alaskan Chinook Salmon

The two variables studied most frequently to explain trends in Alaska Chinook salmon during their ocean residence were the North Pacific Gyre Oscillation (an index of salinity, nutrient upwelling, and surface chlorophyll-a in the Northeast Pacific; Di Lorenzo et al. 2008) and density-dependent competition with pink salmon (*Oncorhynchus gorbuscha*, Salmonidae) (Figure 2). The North Pacific Gyre Oscillation had a positive association with a greater influence on Chinook salmon abundance. Pink salmon had a negative association with a greater effect on Chinook salmon size. Sea surface temperature, at various times of the year, was also relatively well studied, but the evidence was mixed and relatively weaker than other independent variables. The strong negative evidence of marine predation, size-selective harvest, and pink salmon on Chinook salmon size (e.g., size-at-age, mean age, etc.) is notable.

Variables with positive associations can largely be classified as environmental variables, and as such, cannot be directly managed. Conversely, many of the variables with negative associations can be classified as human-created drivers of change, such as bycatch, size-selective harvest, and density-dependent competition with pink salmon (many of which are commercially produced in hatcheries and released into the ocean). It is even suggested that marine predation has increased over time because of the protections placed on marine species (notably the Marine Mammal Protection Act of 1972) that consume Chinook (Seitz et al. 2019; Ohlberger et al. 2019; Chasco et al. 2017).

Many variables have also been examined to explain trends in Alaska Chinook salmon during their time in freshwater, for both juvenile and returning adult salmon (Figure 3). There were relatively few variables with a positive association, and river discharge, smolt size, and incubation temperature were the only variables with positive associations that have been studied more

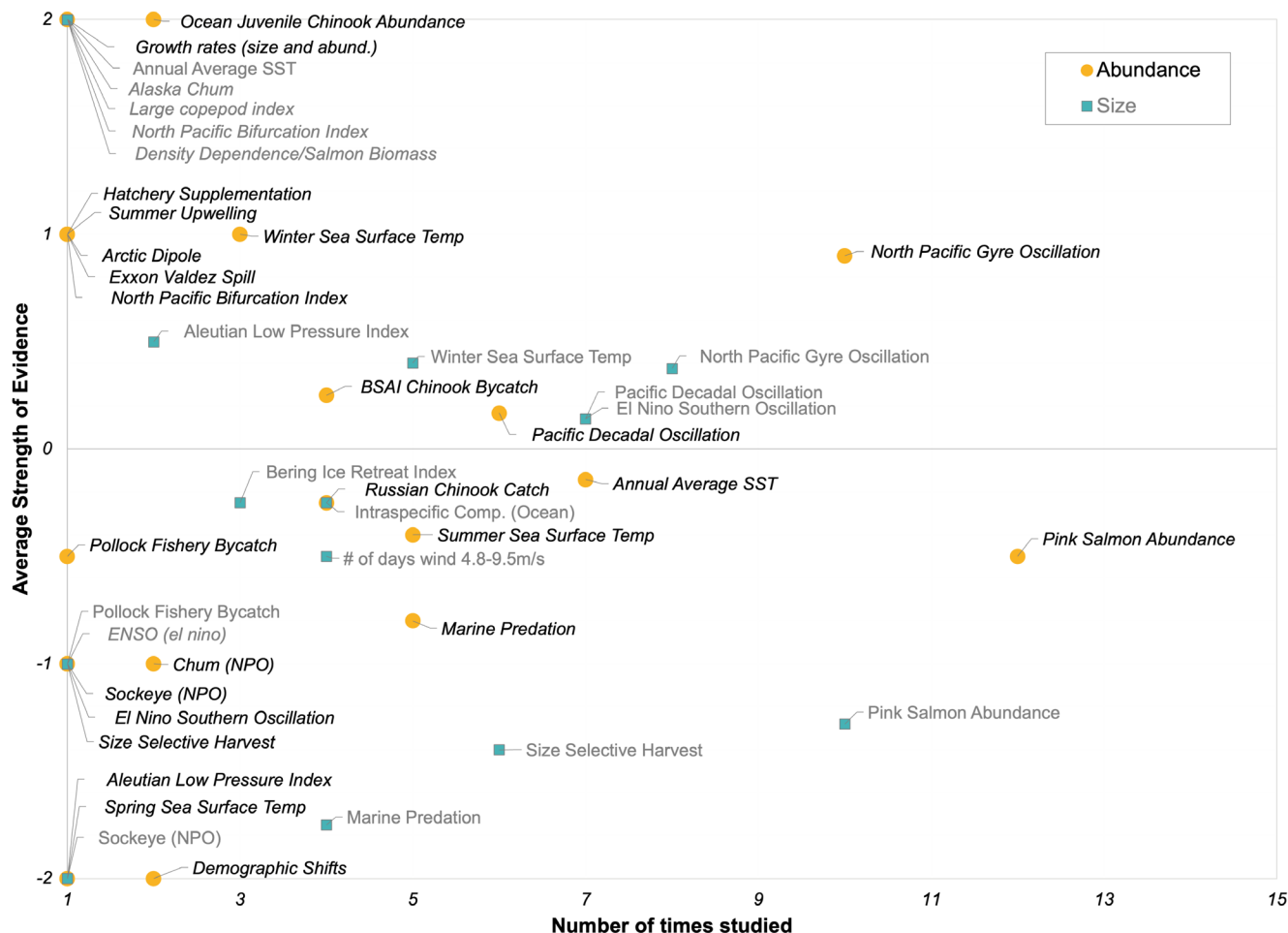


FIGURE 2 | Distribution of ocean variables associated with abundance and size trends in Alaska Chinook salmon. Dependent variables were grouped into “Abundance” (i.e., abundance, productivity, recruits-per-spawner, and survival; black text) or “Size” (i.e., age structure, age-at-maturity, average age, average size, body size, fecundity, growth, maturation, and size-at-age; grey text). All ocean variables with *average strength of significance* = 0 are not shown here but included in Data S3.

than once. Of those variables with negative associations, river ice breakup was studied the most, with survival decreasing with persisting spring river ice (Cunningham et al. 2018; Ohlberger et al. 2016). Elevated river temperature (i.e., migration water temperature) also had a negative association with returning adult Chinook salmon (Murdoch et al. 2024; Howard and von Biela 2023; von Biela et al. 2022; von Biela et al. 2020; Bowen et al. 2020), and freshwater predation had a negative association with juvenile Chinook salmon (Dunker et al. 2020; Sepulveda et al. 2013).

4 | Discussion

Just under 100 independent variables have been explored in analyses attempting to explain trends in Alaska Chinook salmon populations. Most have only been explored in a limited number of studies at localised scales; however, there were variables identified in this synthesis that repeatedly demonstrated strong evidence of association. The results of this synthesis should improve our shared understanding of the complex processes

shaping the future of Chinook salmon, and provide a path forward for research, management, and policy.

4.1 | Drivers for Large-Scale Synchronous Trends

The primary concern for Chinook salmon is the synchronous and large-scale declines in abundance, survival, size-at-age, and mean age. Ultimately, fewer Chinook salmon are returning to spawn, and those that do return are trending younger and smaller, which contributes to a lower reproductive capacity and negative socio-economic effects (Oke et al. 2020). These negative trends have also become more synchronous over time (Dorner et al. 2018).

There is strong evidence suggesting migrating adult heat stress (Bowen et al. 2020; von Biela et al. 2020) and freshwater predation of juveniles (Dunker et al. 2020) negatively impact Chinook salmon. However, Alaska's freshwater habitats are highly diverse, and not all rivers experience these two pressures. Additionally, adult heat stress and juvenile predation are



FIGURE 3 | Distribution of freshwater variables associated with abundance and size trends in Alaska Chinook salmon. Dependent variables were grouped into “Abundance” (i.e., abundance, productivity, recruits-per-spawner, and survival; black text) or “Size” (i.e., age structure, age-at-maturity, average age, average size, body size, fecundity, growth, maturation, and size-at-age; grey text). All freshwater variables with average strength of significance = 0 are not shown here but are included in Data S4.

primarily associated with abundance (Figure 3). Therefore, it is unlikely that freshwater variables are the primary driver for synchronous trends in late-stage survival, size-at-age, and mean age across all regions within the state of Alaska.

The data for heat stress and juvenile predation are very compelling, but over space and time, we would expect these variables to have varying effects. For example, warming stream temperatures on a regional scale have mixed effects depending on the population/river Jones et al. (2020). During warm periods, glacially fed systems have more stable temperature and flow, which results in higher migration success compared to snow/rain driven systems von Biela et al. (2022). With warming, colder rivers are expected to become more productive as temperatures become more optimal, and warmer rivers are expected to become less productive as water temperatures surpass suitable thresholds Shaftel et al. (2020); Jones et al. (2020). Also, warm winters and early spring ice off is associated with improved productivity for juvenile salmon Murdoch et al. (2024). The Kenai River, a glacially fed system, only briefly surpassed 18°C during the drought of 2019 (USGS, Kenai River at Soldotna, AK), yet the

Kenai—and other glacial systems like the Copper River—are experiencing the same negative size and age trends as snow and rain driven systems Lewis et al. (2015).

Furthermore, freshwater predation has had large impacts in watersheds where Northern Pike (*Esox Lucius*, Esocidae) are present and not native (like many rivers flowing into the Cook Inlet). Yet, pike are present and native in many historically productive Chinook salmon systems—like the Yukon River—which are experiencing the same negative trends. Also, freshwater predation only affects juvenile Chinook salmon (abundance), and therefore cannot explain the trends in size and age structure of returning adult Chinook salmon. These, and other freshwater variables can negatively influence Chinook salmon survival, and it is highly likely they are exacerbating negative trends in specific watersheds, which can have a compounding effect for specific populations or regions. However, the common trends observed across populations from spatially distant natal rivers across Alaska suggest the influence of conditions experienced by all populations in their shared ocean environment.

Conditions experienced in the North Pacific Ocean are shared across Chinook salmon stocks from distinct freshwater systems, and therefore could impact all Alaska Chinook populations. The results from this synthesis suggest the North Pacific Gyre Oscillation is a reliable index of Chinook salmon productivity, particularly abundance. This finding supports the conclusion of previous research, which states the North Pacific Gyre Oscillation is the dominant climatic force driving salmon productivity Couture et al. (2024); Kilduff et al. (2015). Results from this synthesis suggest the North Pacific Gyre Oscillation is a better predictor of Chinook salmon productivity when looking at populations over a wide spatial scale. For example, Ohlberger et al. (2016) tested the North Pacific Gyre Oscillation at statewide and regional scales, finding it matched productivity dynamics over large—but not regional—spatial scales better than other indices. Other studies have also found the North Pacific Gyre Oscillation to be strongly associated with Chinook salmon productivity over large spatial scales (e.g., Buckner et al. 2023; Dorner et al. 2018; Kilduff et al. 2015). One exception was Oke et al. (2020) who found little association between the North Pacific Gyre Oscillation and Chinook salmon size structure across Alaska, but did find a strong negative effect on chum and a strong positive effect on coho. Studies using the North Pacific Gyre Oscillation at smaller scales Cunningham et al. (2018) explored Yukon Chinook salmon and Jones et al. (2020) explored Cook Inlet Chinook salmon) found weaker associations. However, one reason Jones et al. (2020) may have found little support is that the North Pacific Gyre Oscillation was tested against the first year of marine residency, and the North Pacific Gyre Oscillation may be a better predictor later in the Chinook life cycle. However, recent research has argued the North Pacific Gyre Oscillation is not as reliable as it was historically for predicting ocean productivity given the decoupling of this index with oceanographic processes Litzow et al. (2020). While interannual and interdecadal climatic variation can impact Chinook salmon productivity, the exact mechanism behind these impacts remains unclear, particularly given the non-stationarity of the relationships underlying the commonly used climate indices Litzow et al. (2020). While the North Pacific Gyre Oscillation currently has predictive power, it is conceivable that the relationship between it and Chinook salmon productivity will deteriorate in the future.

The most pronounced finding from this review is the strength of evidence for marine predation, size selective harvest, and pink salmon abundance on Chinook salmon size (i.e., size-at-age and mean age). The negative association with pink salmon abundance suggests the presence of density-dependent competition for limited food resources in the North Pacific (Murdoch et al. 2024; Ruggerone et al. 2023; Buckner et al. 2023). In addition, as returning Chinook salmon are becoming smaller and younger, on average, the length-at-age of young Chinook salmon has increased (Ohlberger et al. 2018; Lewis et al. 2015). This change may suggest maturation occurring earlier. Interspecies competition between salmon has been shown to affect maturation schedules (Cline et al. 2019). The weight of research also suggests marine predation and size selective harvest are likely contributors to declining Chinook size (Seitz et al. 2019; Ohlberger et al. 2019; Chasco et al. 2017). For example, Chinook populations with greater spatial overlap with coastal populations

of fish-eating whales show the largest declines in size-at-age (Buckner et al. 2023).

There may also be a synergistic effect between two or more of these variables—including freshwater variables. For example, periods of low ocean productivity—like the current sustained low North Pacific Gyre Oscillation—may exacerbate interspecies competition, and species targeting large Chinook salmon could have an oversized effect during these times. In addition, increased salmon abundance from hatcheries could be inflating the number of marine predators that are also effective in targeting large Chinook salmon. Historic, and perhaps even current, harvest methods could also interact with these dynamics even though harvest intensity and methods have varied over space and time (Staton et al. 2021).

Overall, the results of this synthesis suggest the factors associated with large-scale, synchronous reductions in Chinook salmon abundance and size are primarily occurring in the ocean. However, results also suggest freshwater variables like adult heat stress and juvenile predation have affected Chinook salmon (primarily abundance), and can exacerbate negative trends. Furthermore, ocean productivity, specifically the North Pacific Gyre Oscillation, has been the most reliable predictor for Chinook salmon abundance. Marine predation, size-selective harvest, and the direct or indirect effects of increasing pink salmon abundance are, based on the literature, the most likely drivers leading to reduced size and age. While the methods used in this synthesis cannot prove causality, interpretations of causation are appropriate when there is compelling evidence for a given association that has a known foundational mechanism, such as competition. We also acknowledge our analysis is limited to exploring the effects of those variables that have already been examined in the literature. It is possible there are additional influential drivers of Chinook salmon dynamics that have not been considered or for which the time series of data needed to effectively test their impact is unavailable. Despite this limitation, the repeated findings of the effects of North Pacific Gyre Oscillation, marine predation, size-selective harvest, and competition with pink salmon provide strong support for their impacts on Alaska Chinook salmon.

4.2 | Implications for Policy and Management

Evidence suggests large-scale synchronous Chinook salmon trends are being driven by ocean-related factors. Therefore, the next step is to consider what variables can be adjusted, whether directly or indirectly, by policy and/or management to get more and larger Chinook salmon onto spawning grounds. Although we focus on ocean variables below, management and policy should still consider what actionable options may be successful in addressing the impacts from freshwater variables like adult heat stress and juvenile predation. Current efforts, such as removing pike in Cook Inlet (Dunker et al. 2020) and improving Chinook salmon passage in the Yukon River (ADFG – DCF 2023), can reduce pressure on populations as they face other challenges that are more difficult to address.

Variables associated with Chinook salmon trends can generally be managed directly or indirectly. Climatic and oceanic

conditions, like warming water temperatures and the North Pacific Gyre Oscillation, cannot be altered in the short term by state or federal management decisions. As such, management must aim to support populations against the backdrop of variable, changing, and often unpredictable climatic conditions. For example, the North Pacific Gyre Oscillation has been in the lowest sustained phase in recorded history since 2018 (NOAA n.d.). Given this, management decisions can account for known climatic drivers of Chinook population dynamics by implementing adaptive management strategies that are responsive to changing oceanic and climatic conditions.

Regarding density dependence, salmon density in the North Pacific is at an all-time high, and the years 2018, 2019, 2021, and 2023 had the highest salmon abundance on record since 1925 (Connors et al. 2025). Between 1990 and 2015, hatchery salmon accounted for approximately 40% of the total salmon biomass in the ocean (Ruggerone and Irvine 2018). Density dependence is seemingly affecting more than just Chinook salmon, as sea birds (Van Hemert et al. 2020), fish-eating killer whales (Kotik et al. 2023; Ruggerone et al. 2019), and other species of salmon (Rand and Ruggerone 2024; Ohlberger et al. 2023) are all showing signs of nutritional stress. However, not all literature reviewed here detected a negative effect of pink and chum salmon on Chinook salmon. One possible explanation for these inconsistent results could be the independent variable used. For example, despite different methods, studies using the total abundance of pink salmon more consistently found a significant result (Murdoch et al. 2024; Buckner et al. 2023; Oke et al. 2020) than studies using a less comprehensive measure of abundance (e.g., Kamchatka pink salmon, Ohlberger et al. 2016). One exception is Cunningham et al. (2018), where total pink salmon abundance only had a moderate negative effect, and Japanese hatchery chum salmon had a stronger negative effect on Chinook salmon abundance. Where these studies occurred did not consistently affect the outcome of this variable; significant associations between pink salmon abundance and Chinook salmon were identified throughout the state.

Providing some additional nuance to this research, these effects have been observed in odd-even year oscillations that correspond to the two-year lifecycle of pink salmon (Kendall et al. 2020). There is some evidence to suggest pink salmon are having a greater effect on older Chinook salmon (Ruggerone et al. 2016), which could account for the late-stage mortality being observed in Chinook salmon (Manishin et al. 2021; Ohlberger et al. 2018; Lewis et al. 2015). Beyond correlational studies, there is additional evidence suggesting density-dependent interactions between pink and Chinook salmon. Specifically, in the central Bering Strait (52°–58° N), dietary overlap in odd years—when pink salmon were 40 times more abundant—resulted in a 56% reduction in the total weight of prey consumed by Chinook salmon, and only 23% in pink salmon, when compared to even years (Davis et al. 2005). The reduction in prey consumed by Chinook salmon is notable, but so is the greater efficiency of pink salmon predation. There is also a long-term (1952–2021) inverse relationship between Chinook salmon harvest/weight and pink salmon abundance (Ruggerone et al. 2023).

The preponderance of negative relationships between Chinook and pink salmon suggests addressing the hatchery production of pink salmon may be an effective approach for conserving Alaska Chinook salmon. As hatchery operations are entirely subject to human decisions, management can theoretically have a meaningful impact on this apparent driver of Chinook salmon declines by reducing hatchery salmon production. However, as hatchery production also originates in Canada, Russia, and Japan, fully addressing hatchery effects would require international cooperation, a challenging outcome given differing economic, conservation, and management concerns (Connors et al. 2025). The North Pacific Ocean is a common resource; therefore, reductions in just U.S.-based hatchery operations would have a direct and substantial negative impact on Alaska stakeholders, but Alaska stakeholders may not receive the potential benefits of reduced U.S. hatchery production if other resource users (e.g., Canada, Russia, Japan) maintain or increase hatchery production.

Our review found modest support across a relatively large number of studies for a negative relationship between size-selective harvest and Chinook salmon size. Harvest practices within commercial fishing (Staton et al. 2021; Ohlberger et al. 2019; Bromaghin et al. 2011) and recreational angling (Kendall and Quinn 2011), even bycatch (Cunningham et al. 2018), selectively remove larger and older Chinook salmon. Harvest size and gear specifications are the primary methods through which fisheries are managed. These methods intend for the protection of specific size classes of fish to promote sustainability. As such, regulations can be put in place to adjust the size-selectivity of fishing practices, by changing mesh size regulations, allowable gear types, allowable harvest size, or spatio-temporal distribution of fishing pressure. Such regulation changes could, however, produce unintended consequences (e.g., increased bycatch of non-target species), and would therefore need to be monitored closely. It is important to note that fishing gear and fishing pressure have varied over space and time, and research (e.g., Ohlberger et al. 2019) suggests that fishing alone is not capable of explaining the recent widespread synchronous demographic changes in Chinook salmon.

In response to the current collapse of Chinook salmon abundance in the Yukon River, subsistence harvest is now limited to small mesh (<4 in.), which is intended to target whitefishes and other small resident species. Incidental harvest of Chinook salmon is predominantly of small “jack” Chinook salmon that mature younger and smaller than the youngest females in the population. As such, the small mesh is designed to have as little impact on the spawning productivity of the Chinook population while still allowing as much harvest as possible for other available species.

Marine predation also emerged as a likely contributing factor for declining Chinook size and abundance. Across the North Pacific, fish-eating killer whales and pinnipeds have increased in abundance over the last 40 to 50 years (Ohlberger et al. 2019; Chasco et al. 2017; Adams et al. 2016). Various species of salmon-eating sharks (e.g., *Lamna ditropis*, Lamnidae; *Somniosus pacificus*, Somniosidae) have also been identified

as potential drivers for late-stage mortality (Manishin et al. 2021; Courtney et al. 2021; Seitz et al. 2019). Abundance data for these species and evidence of salmon predation is much more uncertain in the Bering Sea, but there are speculations of increased abundance and predation occurring there as well. Diet analysis, simulations, and pop-up satellite archival tags are the three approaches used to explore marine predation, and each has provided evidence for high levels of predation on Chinook salmon that can result in meaningful changes in both abundance and size. For example, diet analysis has shown that fish-eating killer whales (*Orcinus orca*, Delphinidae) consume Chinook salmon throughout the year, with Chinook comprising nearly 100% of their diet in the spring (Hanson et al. 2021). Research has also suggested pinniped predation has the largest impact on abundance, but killer whales target mature Chinook, which account for a greater amount of total Chinook biomass (Chasco et al. 2017). Addressing marine predation is fraught with complex social and ecological considerations since the primary method is culling, which depending on the species under consideration is highly divisive, socially unacceptable, and/or illegal. Many of these considerations also imply conflicting management objectives between multiple species (Ohlberger et al. 2019). However, special congressional approvals have been given for lethally culling pinnipeds on the Columbia River because of their outsized impacts on salmon (Washington Department of Fish and Wildlife n.d.), and managers in Alaska are exploring similar options (personal communication).

The potential trade-offs inherent in every management action associated with the primary variables identified in this review need to be carefully considered in the context of their consequences for fishery-dependent human communities and the ecosystem at large. The geographic range of Chinook salmon complicates the potential for small scale test cases to explore the efficacy of management options. For example, it is likely not possible to remove density-dependent competition and marine predation for a single Chinook population to see if these actions produce positive outcomes. Therefore, it is going to take sophisticated thinking, compromises, and cooperation to address these drivers, if they can be addressed at all.

Given the management options available, and their associated difficulties, we see some potential options moving forward that could benefit multiple species and minimise social and economic impacts. First, if hatchery produced salmon are resulting in meeting, or exceeding, the capacity of the North Pacific Ocean under current environmental conditions, we suggest devising practices that account for climatic indices of productivity (e.g., the North Pacific Gyre Oscillation) when releasing hatchery salmon. Akin to adjusting livestock numbers to reflect fluctuating range conditions, hatchery releases could also account for the variability in ocean productivity to reduce the impact on Chinook salmon and other species. This approach, which is strongly supported by data, is also the most accessible management option. Second, if targeted efforts could reduce marine predation at critical periods (e.g., smolt river exit and adult river entry), there would likely be positive benefits. Ultimately, the goal of management and policy should be to think of ways to reduce density dependent pressures in the ocean, reduce mortality from marine predators (especially late-stage mortality),

and regulate fisheries—all the while accounting for changes in ocean productivity—to get older and larger Chinook to spawning grounds.

4.3 | Future Research

This synthesis provides insights into what research needs immediate and intense focus. First, research, management, and policy could benefit from updated trend studies (e.g., Lewis et al. 2015) that illustrate the type, scale, and synchrony of Chinook salmon trends. It is well understood that these trends have continued to worsen, but updated research and datasets will increase our understanding and allow researchers to further explore the effects of the notable covariates identified in this synthesis. Second, more targeted research on the effects of—and potential interactions between—the North Pacific Gyre Oscillation, marine predation, size-selective harvest, and pink salmon abundance is needed to inform actions to conserve Chinook salmon moving forward.

Negative publication bias occurs when research with statistically significant findings is published at a disproportionately higher rate than research with no significant findings. However, given the multivariate nature of Chinook salmon research, there are many variables that continually get published with little or no relationship to Chinook salmon trends. The body of literature included in this review has evaluated the effect of nearly 100 independent variables on Chinook salmon, many of which repeatedly produced non-significant findings. We have listed all 35 “non-significant” variables in tables (Data S3 and S4). Some of these variables have only been used in one analysis and warrant further investigation. Others, such as Arctic Oscillation, have been included in up to five different analyses, and their overall significance is nil. Therefore, we hope this synthesis helps prioritise less studied variables with potentially meaningful findings and deprioritise others that have been repeatedly studied and produced no significant results.

This synthesis has also shown that salmon research is constrained by the availability of data, which creates bias. Specific covariates are repeatedly incorporated into scientific analyses because of their consistent collection, ease of interpretation, and ease of access. For example, freshwater ice breakup, Pacific Decadal Oscillation, North Pacific Gyre Oscillation—among many others—fall into this category. Data on prey availability and abundance in the North Pacific Ocean are difficult and expensive to collect, especially at the scale necessary to make inferences about their impacts on salmon (although these efforts do exist—e.g., Murphy et al. 2023). The potential bias is hard to assess but may result in overemphasis of certain well-studied variables and their effects on salmon.

When exploring a new salmon system, we—rightly so—rely on the past literature to guide hypothesis development and are subject to the availability of time series of environmental covariates. This process, however, prevents the exploration of new hypotheses. We hope this synthesis redirects and focuses research to better understand the relationships between Chinook salmon and the key variables we have identified, including the interactions

between these key variables. We also hope this synthesis provides insights into new hypotheses and variables. Moving forward, researchers need to be thoughtful about what processes or sources of knowledge/information are not well represented in the current scientific literature but have the potential to be informative.

In conclusion, this synthesis has identified likely drivers for the alarming and impactful trends of Chinook salmon across Alaska. We hope this information fosters a shared understanding of the complex processes affecting Chinook salmon and increases the likelihood that invested parties can move collectively toward a course of action(s) that will improve our understanding and the sustainability of this culturally and economically vital species.

Author Contributions

Chase C. Lamborn: conceptualisation, formal analysis, funding acquisition, investigation, methodology, project administration, supervision, validation, visualisation, writing – original draft. **Jan Ohlberger:** conceptualisation, validation, writing – original draft. **Timothy E. Walsworth:** conceptualisation, validation, writing – original draft. **Peter A. H. Westley:** conceptualisation, writing – original draft. **Curry J. Cunningham:** conceptualisation, writing – original draft. **Sam Wynsma:** investigation, validation, visualisation, writing – original draft. **Jordan W. Smith:** conceptualisation, funding acquisition, project administration, supervision, visualisation, writing – original draft.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The spreadsheet containing all the literature incorporated into this synthesis—which includes the geography, life stage, dependent and independent variables used in each analysis, and the significance of each independent variable—is available as [Supporting Information](#).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.